

Ticket #007: DNS Resolution Failure

Symptoms: The server is experiencing failures resolving domain names while maintaining general network connectivity. This is impacting outbound connections to external services that rely on hostname resolution.

Resolution:

1. Verifying the server's general network connectivity.

```
[jay@mainserver:~$ ping -c 2 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=115 time=32.5 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=115 time=25.5 ms

--- 8.8.8.8 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1002ms
rtt_min/avg/max/mdev = 25.522/29.002/32.483/3.480 ms
```

Analysis: The server is maintaining network connectivity.

2. Verifying DNS resolution.

```
[jay@main-server:~$ nslookup google.com
;; communications error to 127.0.0.53#53: connection refused
;; communications error to 127.0.0.53#53: connection refused
;; communications error to 127.0.0.53#53: connection refused
;; no servers could be reached
```

Analysis: DNS queries are directed to the local stub resolver at 127.0.0.53; however, the connection is actively refused because the stub listener is not available. As a result, no upstream DNS queries are initiated.

3. Verifying whether the resolver service is running.

```
[jay@main-server:~$ sudo systemctl status systemd-resolved
● systemd-resolved.service - Network Name Resolution
   Loaded: loaded (/usr/lib/systemd/system/systemd-resolved.service; enabled; preset: enabled)
   Active: active (running) since Mon 2026-05-04 18:41:59 UTC; 1min 49s ago
     Docs: man:systemd-resolved.service(8)
           man:org.freedesktop.resolve1(5)
           https://www.freedesktop.org/wiki/Software/systemd/writing-network-configuration-managers
           https://www.freedesktop.org/wiki/Software/systemd/writing-resolver-clients
   Main PID: 1040 (systemd-resolve)
   Status: "Processing requests..."
     Tasks: 1 (limit: 4544)
    Memory: 2.5M (peak: 2.7M)
       CPU: 23ms
   CGroup: /system.slice/systemd-resolved.service
           └─1040 /usr/lib/systemd/systemd-resolved
```

Analysis: The service is active and running.

4. Verifying which service is bound to port 53.

```
[jay@mainserver:~$ ss -tulnp | grep :53
jay@mainserver:~$
```

Analysis: DNSStubListener is not bound to port 53 despite the systemd-resolved service being active.

5. Verifying whether the DNSStubListener is enabled.

```
[jay@mainserver:~$ cat /etc/systemd/resolved.conf
```

```
#Domains=
#DNSSEC=no
#DNSOverTLS=no
#MulticastDNS=no
#LLMNR=no
#Cache=no-negative
#CacheFromLocalhost=no
DNSStubListener=no
#DNSStubListenerExtra=
#ReadEtcHosts=yes
#ResolveUnicastSingleLabel=no
#StaleRetentionSec=0
```

Analysis: The resolved.conf file shows DNSStubListener is set to no, indicating the stub listener is disabled.

6. Updating DNSStubListener from no to yes in resolved.conf.

```
#Domains=
#DNSSEC=no
#DNSOverTLS=no
#MulticastDNS=no
#LLMNR=no
#Cache=no-negative
#CacheFromLocalhost=no
DNSStubListener=yes
#DNSStubListenerExtra=
#ReadEtcHosts=yes
#ResolveUnicastSingleLabel=no
```

7. Restarting systemd-resolved and verifying whether the service is bound to port 53.

```
[jay@main-server:~$ sudo ss -tulnp | grep :53
[jay@main-server:~$ sudo nano /etc/systemd/resolved.conf
[jay@main-server:~$ sudo systemctl restart systemd-resolved
[jay@main-server:~$ sudo ss -tulnp | grep :53
udp UNCONN 0      0                127.0.0.54:53    0.0.0.0:*        users:(("systemd-resolve",pid=1340,fd=16))
udp UNCONN 0      0                127.0.0.53%lo:53 0.0.0.0:*        users:(("systemd-resolve",pid=1340,fd=14))
tcp LISTEN 0      4096            127.0.0.53%lo:53 0.0.0.0:*        users:(("systemd-resolve",pid=1340,fd=15))
tcp LISTEN 0      4096            127.0.0.54:53    0.0.0.0:*        users:(("systemd-resolve",pid=1340,fd=17))
```

Analysis: Systemd-resolved successfully binds to port 53

8. Testing DNS resolution.

```
jay@mainserver:~$ nslookup google.com
Server:      127.0.0.53
Address:     127.0.0.53#53

Non-authoritative answer:
Name:   google.com
Address: 192.178.155.113
Name:   google.com
Address: 192.178.155.138
```

Analysis: DNS queries are successful. The Issue is fully resolved.

Conclusion: The DNS resolution failure was caused by the systemd-resolved stub listener being disabled, preventing the local resolver at 127.0.0.53 from accepting queries. Although the resolver service was running, it was not bound to port 53, which blocked all upstream DNS lookups. Enabling the stub listener and restarting systemd-resolved restored proper binding and re-established normal DNS functionality, fully resolving the issue.